[MMS 2025/26] PROJECT: IMAGE COMPRESSION METHODS

Total 5 points: Implementation (GitHub repository) 2p + Documentation 2p + Presentation 1p

REQUIREMENTS

Run (modify) the provided code on the ImageNet dataset and present results.
Note: Each topic can be taken by one student or a pair of 2 students. Notify me if you need more topics. Share documents with me: vhasic1@etf.unsa.ba

Implementation (2p)

Write PyTorch code to run the chosen algorithm on the ImageNet-1k dataset. Every paper listed below has code provided (you only need to slightly modify it to run on ImageNet). Evaluate the code with same metrics as the authors did in the paper. Notify me if a paper does not provide code.

Documentation (2p)

Max 10 pages. Must include:

- What problem does the paper tackle
- Explain the algorithm (you need to understand the algorithm)
- Explain your implementation
- Explain the metrics and their meaning
- Detailed descriptions of problems faced and how they were solved
- Present results and their interpretation
- How could the algorithm be improved

Presentation (1p)

Max 10 slides. Must cover:

- Present the key things for every point in the documentation.

TOPICS

1. Ultra-Low Bitrate Perceptual Image Compression with Shallow Encoder
https://arxiv.org/pdf/2512.12229v2

Student/s: Benjamin Hadzihasanovic, Tarik Čaluk
Link to GitHub implementation:

Link to documentation:
Link to presentation:

2. Few-Shot Domain Adaptation for Learned Image Compression
https://arxiv.org/pdf/2409.11111

Student/s: Emina Šahbaz, Edna Džinić, Zehra Buza
Link to GitHub implementation:
Link to documentation:
Link to presentation:

3. Learned Image Compression with Dictionary-based Entropy Model
https://openaccess.thecvf.com/content/CVPR2025/papers/Lu_Learned_Image_Compression_with_Dictionary-based_Entropy_Model_CVPR_2025_paper.pdf

Student/s: Hamza Hrnjic i Safet Comor
Link to GitHub implementation:
Link to documentation:
Link to presentation:

4. Towards Image Compression with Perfect Realism at Ultra-Low Bitrates
https://openreview.net/pdf?id=ktdETU9JBg

Student/s: Edis Jasarevic, Vedad Hajric
Link to GitHub implementation:
Link to documentation:
Link to presentation:

5. Correcting Diffusion-Based Perceptual Image Compression
https://arxiv.org/pdf/2404.04916

Student/s: Amila Kukić, Una Hodžić
Link to GitHub implementation:
Link to documentation:
Link to presentation:

6. StableCodec: Taming One-Step Diffusion for Extreme Image Compression
https://openaccess.thecvf.com/content/ICCV2025/papers/Zhang_StableCodec_Taming_One-Step_Diffusion_for_Extreme_Image_Compression_ICCV_2025_paper.pdf

Student/s: Dženan Devedžić

Link to GitHub implementation:
Link to documentation:
Link to presentation:

7. OSCAR: One-Step Diffusion Codec Across Multiple Bit-rates
https://openreview.net/pdf?id=uodE9CAXaF

Student/s: Eldar Buzadžić, Edi Đelmo
Link to GitHub implementation:
Link to documentation:
Link to presentation:

8. DiffO: Single-step Diffusion for Image Compression
https://openaccess.thecvf.com/content/WACV2026/papers/Park_Single-step_Diffusion_for_Image_Compression_at_Ultra-Low_Bitrates_WACV_2026_paper.pdf

Student/s: Gičević Ajša, Elma Tirak
Link to GitHub implementation:
Link to documentation:
Link to presentation:

9. DiffPC: Diffusion-based High Perceptual Fidelity Image Compression with Semantic Refinement
https://openreview.net/pdf?id=RL7PycCtAO

Student/s: Amer Mujalo, Dzeneta Milic, Lamija Gutic
Link to GitHub implementation:
Link to documentation:
Link to presentation:

10. HDCompression: Hybrid-Diffusion Image Compression for Ultra-Low Bitrates
https://arxiv.org/pdf/2502.07160

Student/s: Haris Silajdzic, Adil Tutun
Link to GitHub implementation:
Link to documentation:
Link to presentation:

11. Compressed Image Generation with Denoising Diffusion Codebook Models
https://openreview.net/pdf?id=cQHwUckohW

Student/s: Mina Duranović, Sara Dautbegović
Link to GitHub implementation:
Link to documentation:

Link to presentation:

12. CoD: A Diffusion Foundation Model for Image Compression
https://arxiv.org/pdf/2511.18706v1

Student/s: Nejra Pirija, Nađa Kovačević
Link to GitHub implementation:
Link to documentation:
Link to presentation:

13. CADC: Content Adaptive Diffusion-Based Generative Image Compression
https://arxiv.org/pdf/2602.21591v2

Student/s: Azra Smajović, Amina Rokša
Link to GitHub implementation:
Link to documentation:
Link to presentation:

14. One-Step Diffusion-Based Image Compression with Semantic Distillation
https://openreview.net/pdf/18a2e64660dd441138f1c1cc3aac0a683e644a53.pdf

Student/s:
Link to GitHub implementation:
Link to documentation:
Link to presentation:

15. Switchable Token-Specific Codebook Quantization For Face Image Compression
https://openreview.net/pdf?id=8AtYSW3VdX

Student/s:
Link to GitHub implementation:
Link to documentation:
Link to presentation:

16. Differentiable Vector Quantization for Rate-Distortion Optimization of Generative Image Compression
https://arxiv.org/pdf/2604.10546

Student/s:
Link to GitHub implementation:
Link to documentation:
Link to presentation:

17. A lightweight model for perceptual image compression via implicit priors https://www.sciencedirect.com/science/article/pii/S0893608025011608?via%3Dihub
18. Hierarchical Semantic Compression for Consistent Image Semantic Restoration https://arxiv.org/pdf/2502.16799
19. ProGIC: Progressive and Lightweight Generative Image Compression with Residual Vector Quantization https://arxiv.org/pdf/2603.02897

Student/s: Naida Hasović, Lana Malinov
Link to GitHub implementation:
Link to documentation:
Link to presentation:

20. Differentiable Vector Quantization for Rate-Distortion Optimization of Generative Image Compression https://arxiv.org/pdf/2604.10546
21. CADC: Content Adaptive Diffusion-Based Generative Image Compression https://arxiv.org/pdf/2602.21591
22. PerCoV2: Improved Ultra-Low Bit-Rate Perceptual Image Compression with Implicit Hierarchical Masked Image Modeling https://arxiv.org/pdf/2503.09368
23. https://arxiv.org/pdf/2512.20194
24. https://arxiv.org/pdf/2404.18820
25. https://arxiv.org/pdf/2410.02640
26. https://openreview.net/pdf?id=0TSAIUCwpp
27. https://openreview.net/pdf?id=raUnLe0Z04